Task related to ECS.

1. Setup a highly available ecs cluster with load balancer and dynamic port mapping.

NOTE:
Your cluster should maintain atleast 6 tasks at any point of time and should be highly availble across multiple AZ's.
Use the below image from deployment:
sabair0509/hiring-app:works

1. Objective

To deploy a highly available containerized application using Amazon ECS with EC2 launch capacity, an Application Load Balancer (ALB), and dynamic port mapping. The service should maintain at least six running tasks across multiple Availability Zones and distribute incoming traffic to healthy application containers.

2. Steps used

Step 1 — Create the ECS cluster

- Opened Amazon ECS in the `us-east-1` region.
- Created the cluster `hiring-app-cluster`.
- Configured EC2 capacity for the ECS cluster.

Step 2 — Configure EC2 capacity

- Used two EC2 instances of type `c7i-flex.large`.
- Distributed the instances across:
  - `us-east-1a`
  - `us-east-1b`
- Configured the Auto Scaling Group and ECS capacity provider to supply container-hosting capacity.

Step 3 — Create the task definition

Created task definition `hiring-app-task` and deployed image:

`sabair0509/hiring-app:works`

Configured the container with bridge networking and dynamic host-port mapping. CloudWatch logging was added to help troubleshoot the application.

The application logs confirmed that Tomcat starts on container port `8080` and deploys `hiring.war`.

Step 4 — Create the Application Load Balancer

- Created the internet-facing ALB `hiring-app-alb`.
- Enabled two Availability Zones and selected their subnets.
- Created the target group `hiring-app-tg`.
- Configured an HTTP listener on port `80`.
- Set the listener's default action to forward traffic to `hiring-app-tg`.

Step 5 — Configure dynamic port mapping

Configured the ECS task definition to use container port `8080` with dynamic host ports.

For example:

EC2 host port	Container port
32780	8080
32781	8080
32782	8080
32783	8080

32784	8080
32785	8080

These are example host-port registrations observed during deployment; ECS assigns host ports dynamically.

Step 6 — Configure the ECS service

Created `hiring-app-service` in `hiring-app-cluster` and configured:

- Desired task count: `6`
- Task definition: `hiring-app-task:3`
- Load balancer: `hiring-app-alb`
- Target group: `hiring-app-tg`
- Container port: `8080`
- Health-check grace period: 300 seconds

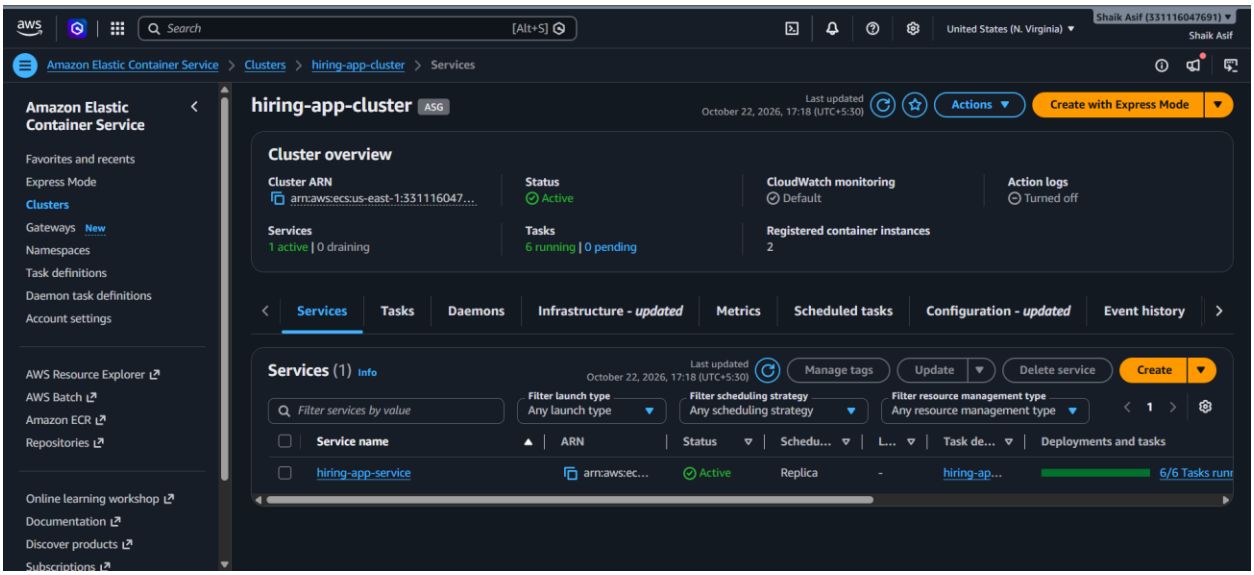
Waited for the deployment and target health checks to stabilize.

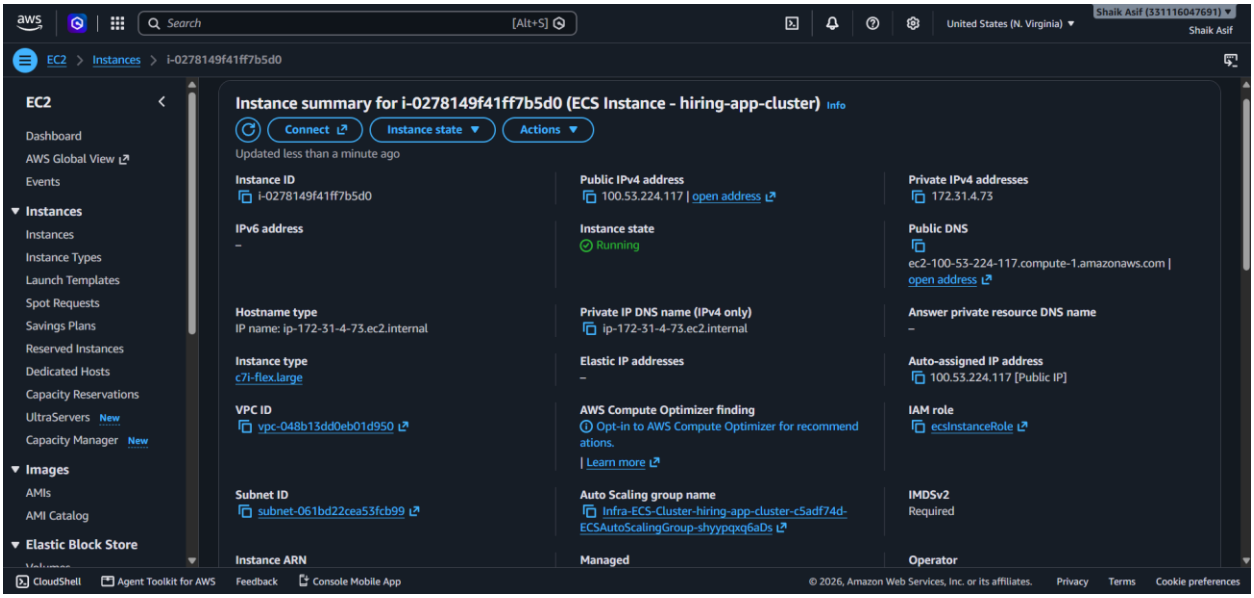
Step 7 — Verify the application

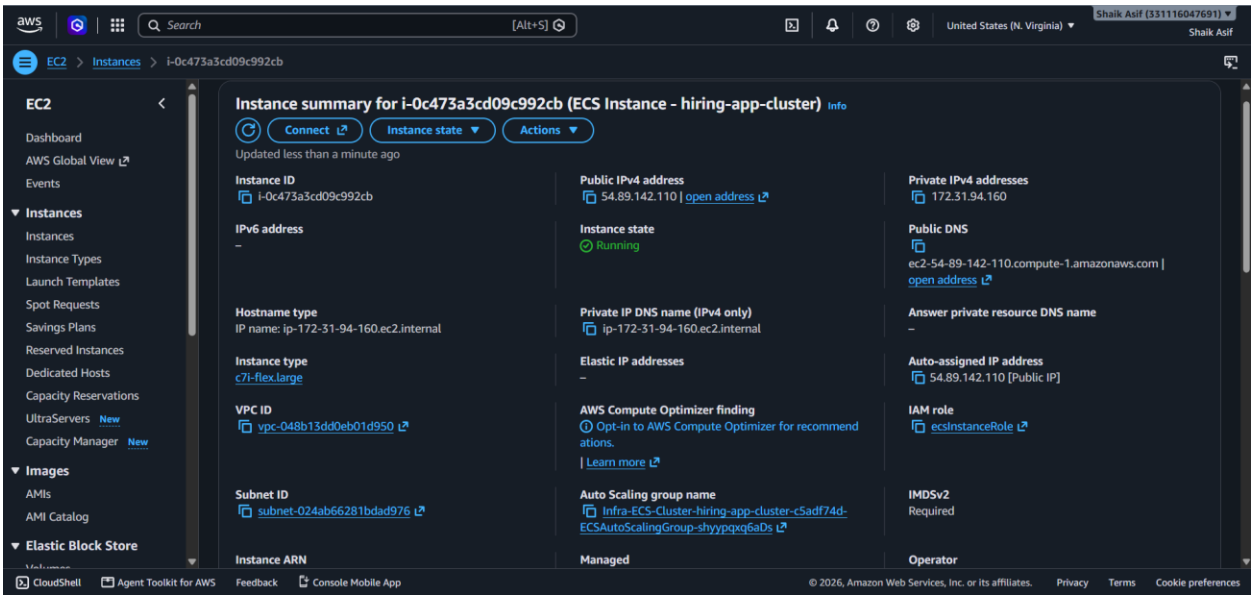
Opened the ALB DNS name in a browser and accessed:

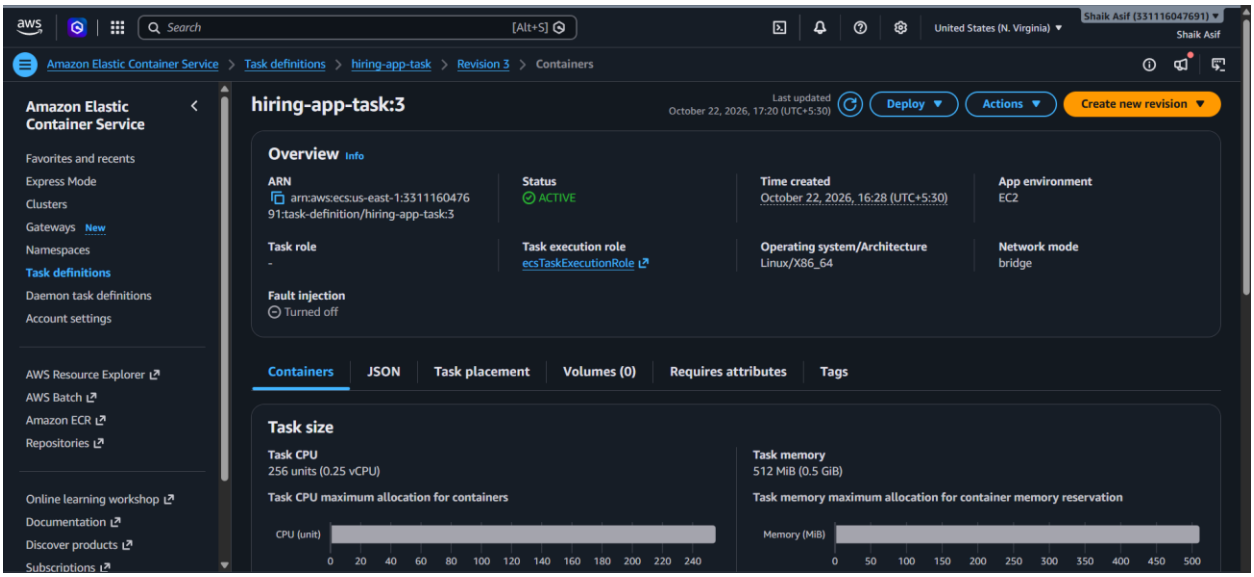
http://hiring-app-alb-2114098881.us-east-1.elb.amazonaws.com/hiring/

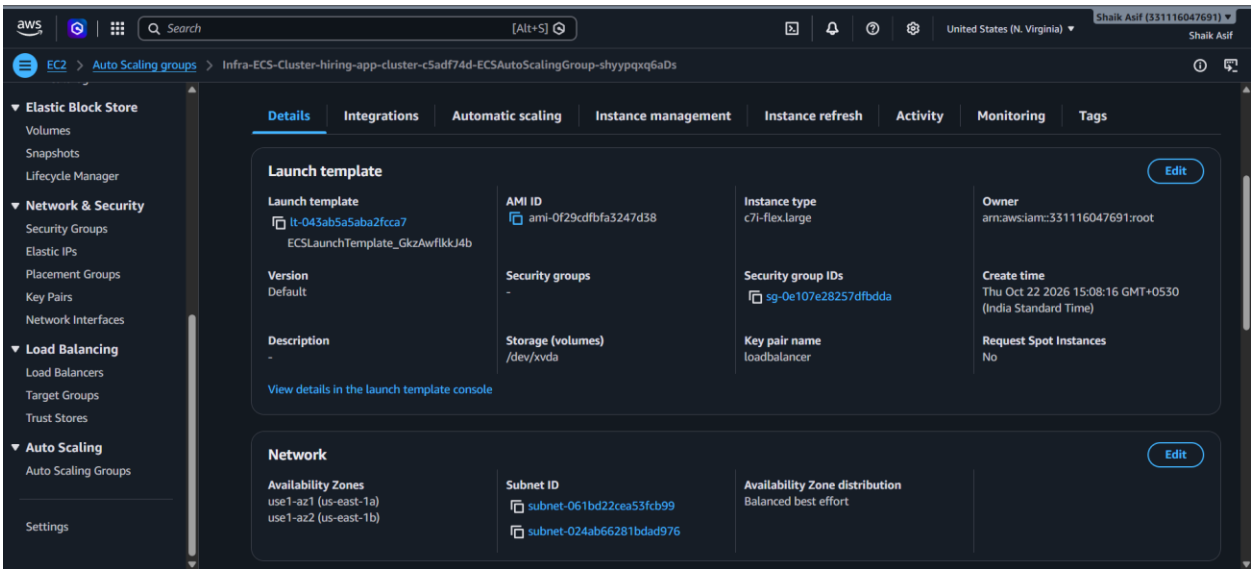
The application displayed “Welcome to Techie Horizon”, confirming that the application was reachable through the load balancer.

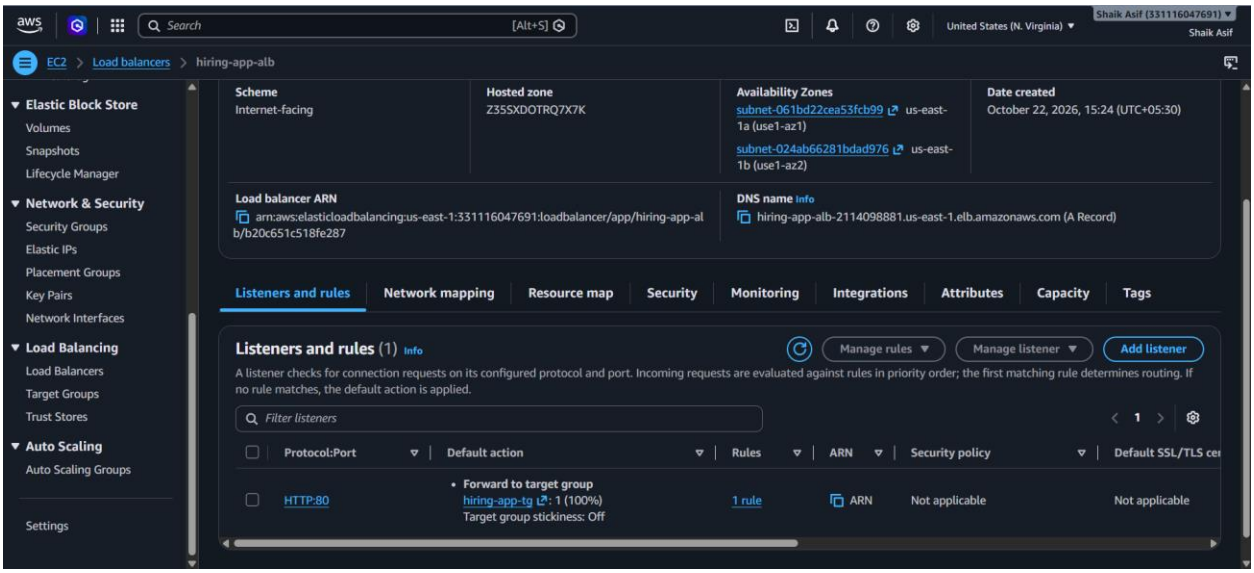

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Namespaces
Task definitions
Daemon task definitions
Account settings
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Actions
Create with Express Mode
Cluster overview
Cluster ARN
arn:aws:ecs:us-east-1:331116047...
Status
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CloudWatch monitoring
Default
Action logs
Turned off
Services
1 active | 0 draining
Tasks
6 running | 0 pending
Registered container instances
2
Services Tasks Daemons Infrastructure - updated Metrics Scheduled tasks Configuration - updated Event history
Services (1) Info
Last updated October 22, 2026, 17:18 (UTC+5:30)
Manage tags
Update
Delete service
Create
Filter services by value
Filter launch type
Any launch type
Filter scheduling strategy
Any scheduling strategy
Filter resource management type
Any resource management type
Service name
ARN
Status
Schedu...
L...
Task de...
Deployments and tasks
hiring-app-service
arn:aws:ec...
Active
Replica
-
hiring-ap...
6/6 Tasks runn

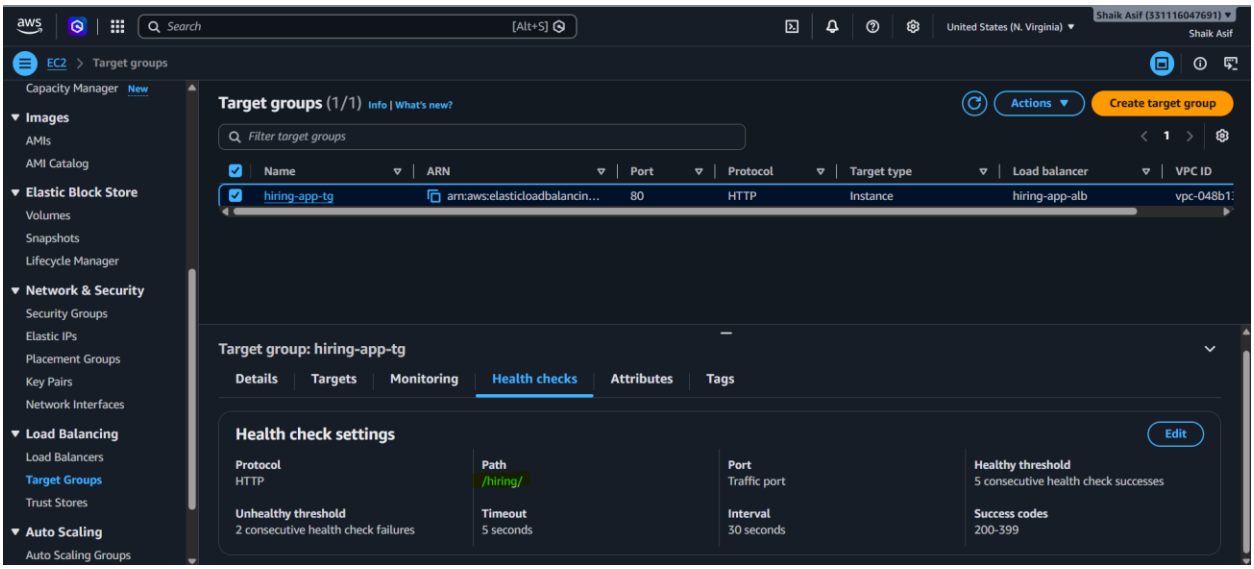

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Events
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Instances
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Reserved Instances
Dedicated Hosts
Capacity Reservations
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Capacity Manager New
Images
AMIs
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Elastic Block Store
Instance summary for i-0278149f41ff7b5d0 (ECS Instance - hiring-app-cluster) Info
Connect
Instance state
Actions
Updated less than a minute ago
Instance ID
i-0278149f41ff7b5d0
Public IPv4 address
100.53.224.117 | open address
Private IPv4 addresses
172.31.4.73
IPv6 address
-
Instance state
Running
Public DNS
ec2-100-53-224-117.compute-1.amazonaws.com | open address
Hostname type
IP name: ip-172-31-4-73.ec2.internal
Private IP DNS name (IPv4 only)
ip-172-31-4-73.ec2.internal
Answer private resource DNS name
-
Instance type
c7i-flex.large
Elastic IP addresses
-
Auto-assigned IP address
100.53.224.117 [Public IP]
VPC ID
vpc-048b13dd0eb01d950
AWS Compute Optimizer finding
Opt-in to AWS Compute Optimizer for recommendations. | Learn more
IAM role
ecsInstanceRole
Subnet ID
subnet-061bd22cea53fcb99
Auto Scaling group name
Infra-ECS-Cluster-hiring-app-cluster-c5adf74d-ECSAutoScalingGroup-shyypqxq6aDs
IMDSv2
Required
Instance ARN
Managed
Operator
CloudShell
Agent Toolkit for AWS
Feedback
Console Mobile App
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UltraServers New
Capacity Manager New
Images
AMIs
AMI Catalog
Elastic Block Store
Instance summary for i-0c473a3cd09c992cb (ECS Instance - hiring-app-cluster) Info
Connect
Instance state
Actions
Updated less than a minute ago
Instance ID
i-0c473a3cd09c992cb
Public IPv4 address
54.89.142.110 | open address
Private IPv4 addresses
172.31.94.160
IPv6 address
–
Instance state
Running
Public DNS
ec2-54-89-142-110.compute-1.amazonaws.com | open address
Hostname type
IP name: ip-172-31-94-160.ec2.internal
Private IP DNS name (IPv4 only)
ip-172-31-94-160.ec2.internal
Answer private resource DNS name
–
Instance type
c7i-flex.large
Elastic IP addresses
–
Auto-assigned IP address
54.89.142.110 [Public IP]
VPC ID
vpc-048b13dd0eb01d950
AWS Compute Optimizer finding
Opt-in to AWS Compute Optimizer for recommendations.
| Learn more
IAM role
ecsInstanceRole
Subnet ID
subnet-024ab66281bdad976
Auto Scaling group name
Infra-ECS-Cluster-hiring-app-cluster-c5adf74d-ECSAutoScalingGroup-shyypqxq6aDs
IMDSv2
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hiring-app-task:3
Last updated
October 22, 2026, 17:20 (UTC+5:30)
Deploy
Actions
Create new revision
Overview Info
ARN
arn:aws:ecs:us-east-1:331116047691:task-definition/hiring-app-task:3
Status
ACTIVE
Time created
October 22, 2026, 16:28 (UTC+5:30)
App environment
EC2
Task role
-
Task execution role
ecsTaskExecutionRole
Operating system/Architecture
Linux/X86_64
Network mode
bridge
Fault injection
Turned off
Containers
JSON
Task placement
Volumes (0)
Requires attributes
Tags
Task size
Task CPU
256 units (0.25 vCPU)
Task memory
512 MiB (0.5 GiB)
Task CPU maximum allocation for containers
CPU (unit)
0 20 40 60 80 100 120 140 160 180 200 220 240
Task memory maximum allocation for container memory reservation
Memory (MiB)
0 50 100 150 200 250 300 350 400 450 500


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Elastic Block Store
Volumes
Snapshots
Lifecycle Manager
Network & Security
Security Groups
Elastic IPs
Placement Groups
Key Pairs
Network Interfaces
Load Balancing
Load Balancers
Target Groups
Trust Stores
Auto Scaling
Auto Scaling Groups
Settings
Details
Integrations
Automatic scaling
Instance management
Instance refresh
Activity
Monitoring
Tags
Launch template
Edit
Launch template
lt-043ab5a5aba2fcca7
ECSLaunchTemplate_GkzAwflkkJ4b
AMI ID
ami-0f29cdfbfa3247d38
Instance type
c7i-flex.large
Owner
arn:aws:iam::331116047691:root
Version
Default
Security groups
-
Security group IDs
sg-0e107e28257dfbdda
Create time
Thu Oct 22 2026 15:08:16 GMT+0530 (India Standard Time)
Description
-
Storage (volumes)
/dev/xvda
Key pair name
loadbalancer
Request Spot Instances
No
View details in the launch template console
Network
Edit
Availability Zones
use1-az1 (us-east-1a)
use1-az2 (us-east-1b)
Subnet ID
subnet-061bd22cea53fcb99
subnet-024ab66281bdad976
Availability Zone distribution
Balanced best effort


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EC2 > Load balancers > hiring-app-alb
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Network & Security
Security Groups
Elastic IPs
Placement Groups
Key Pairs
Network Interfaces
Load Balancing
Load Balancers
Target Groups
Trust Stores
Auto Scaling
Auto Scaling Groups
Settings
Scheme
Internet-facing
Hosted zone
Z35SXDOTRQ7X7K
Availability Zones
subnet-061bd22cea53fcb99 us-east-1a (use1-az1)
subnet-024ab66281bdad976 us-east-1b (use1-az2)
Date created
October 22, 2026, 15:24 (UTC+05:30)
Load balancer ARN
arn:aws:elasticloadbalancing:us-east-1:331116047691:loadbalancer/app/hiring-app-alb/b20c651c518fe287
DNS name Info
hiring-app-alb-2114098881.us-east-1.elb.amazonaws.com (A Record)
Listeners and rules
Network mapping
Resource map
Security
Monitoring
Integrations
Attributes
Capacity
Tags
Listeners and rules (1) Info
Manage rules
Manage listener
Add listener
A listener checks for connection requests on its configured protocol and port. Incoming requests are evaluated against rules in priority order; the first matching rule determines routing. If no rule matches, the default action is applied.
Filter listeners
Protocol:Port
Default action
Rules
ARN
Security policy
Default SSL/TLS cer
HTTP:80
Forward to target group
hiring-app-tg: 1 (100%)
Target group stickiness: Off
1 rule
ARN
Not applicable
Not applicable





Welcome to Techie Horizon

3. Observations

The final ECS service page showed the following results.

Parameter	Final observation
ECS cluster	`hiring-app-cluster`
ECS service	`hiring-app-service`
Desired tasks	6
Running tasks	6

Pending tasks	0
Healthy targets	6
Unhealthy targets	0
Deployment status	Success
Availability Zones	`us-east-1a`, `us-east-1b`
Load balancer	`hiring-app-alb`
Target group	`hiring-app-tg`
Application response	Welcome to Techie Horizon

The final service status confirms that all six tasks were running and all six registered healthy targets were passing the target group's health checks.

4. Troubleshooting performed

Issue 1: Incorrect application port

Problem: Health checks failed because the task definition initially used container port `80`, while the Tomcat application was listening on `8080`.

Fix: Updated the task definition to use container port `8080` and dynamic host-port mapping.

Issue 2: HTTP 302 health-check failures

Problem: The target group reported `Health checks failed with these codes: [302]`.

Fix: Changed the health-check path to `/hiring/` and changed the success-code matcher from `200` to `200-399`, allowing the redirect response to pass the check.

Issue 3: Old targets in Draining state

Problem: The target group temporarily showed 12 registrations, including old targets being removed and new targets being registered.

Fix: Waited for ECS to replace old registrations and verified the final healthy target count.

Issue 4: ALB browser timeout

Problem: The browser showed `ERR_CONNECTION_TIMED_OUT`, even though the targets were healthy.

Investigation: Checked the ALB scheme and state, listener, subnet routes, Internet Gateway, network ACL, and security-group rules.

Root cause: The ALB security group's HTTP port 80 rule referenced the same security group instead of allowing public HTTP traffic.

Fix: Replaced the HTTP rule with a new inbound rule allowing TCP port `80` from `0.0.0.0/0`. The application then became accessible.

5. Conclusion

The high-availability ECS deployment was completed successfully. The application was deployed using Amazon ECS with EC2 capacity, and an internet-facing Application Load Balancer was configured to distribute requests through a target group using dynamic host-port mapping.

The final service status showed six desired and running tasks, zero pending tasks, six healthy targets, and a successful deployment. The application was accessible through the ALB DNS name. The deployment demonstrates container orchestration, load balancing, health checks, dynamic port mapping, multi-AZ placement, and troubleshooting of application and network connectivity.